

Chapter 1

Introduction

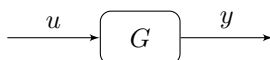
In controls we make use of the abstract concept of a *system*: we identify a phenomenon or a process, the *system*, and two classes of *signals*, which we label as *inputs* and *outputs*. A signal is something that can be measured or quantified. In this book we use real numbers to quantify signals. The classification of a particular signal as an input means that it can be identified as the *cause* of a particular system behavior, whereas an output signal is seen as the *product* or *consequence* of the behavior. Of course the classification of a phenomenon as a system and the labeling of input and output signals is an abstract construction. A mathematical description of a system and its signals is what constitutes a *model*. The entire abstract construction, and not only the equations that we will later associate with particular signals and systems, is the model.

We often represent the relationship between a system and its input and output signals in the form of a *block-diagram*, such as the one in Fig. 1.1. The diagram indicates that a system, G , produces an output signal, y , in the presence of the input signal, u . Block diagrams will be used to represent the interconnection of systems and even algorithms.

1.1 Models and Experiments

System, signals and models are often associated with concrete or abstract experiments. A model reflects a particular setup in which the outputs are correlated with a prescribed set of inputs. For example, we might attempt to model a car

Fig. 1.1: System represented as a block-diagram; u is the input signal; y is the output signal; y and u are related through $y = G(u)$ or $y = Gu$



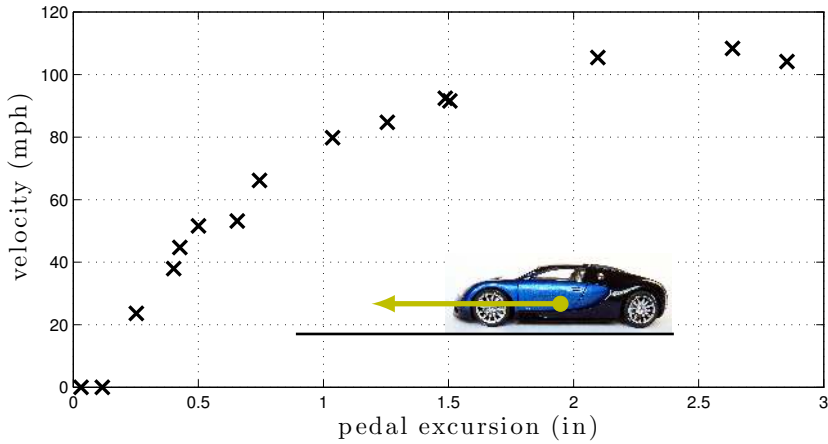


Fig. 1.2: Experimental determination of the effect of the pedal excursion (input, u) on a car's terminal velocity (output, y)

by performing the following experiment: in an unobstructed and level road, we depress the accelerator pedal and let the car travel in a straight line¹. We keep the pedal excursion constant and let the car reach constant velocity. We record the amount the pedal has been depressed and the car's terminal velocity. The results of this experiment, repeated multiple times with different amounts of pedal excursion, might look like the data shown in Fig. 1.2. In this experiment the signals are:

input: pedal excursion, in cm, in, etc;

output: terminal velocity of the car, in m/s, mph, etc.

The **system** is the car *and* the particular conditions of the experiment. Note that the data captures the fact that the car does not move at all for small pedal excursions and that the terminal velocity *saturates* as the pedal reaches the end of its excursion range.

From the data in Fig. 1.2, one might try to *fit* a particular mathematical function to the experimental data in hope of obtaining a *mathematical model*. In doing so, one should be prepared to loose something in name of a simpler description, often the power to describe other behaviors which can not be capture by an equation. Such trade-offs are common place in science, and it should be no different in the analysis and design of control systems. Fig. 1.3 shows the result of fitting a curve of the form

$$y = \alpha \tan^{-1}(\beta u)$$

¹This may bring to memory a bad joke about physicists and spherical cows...

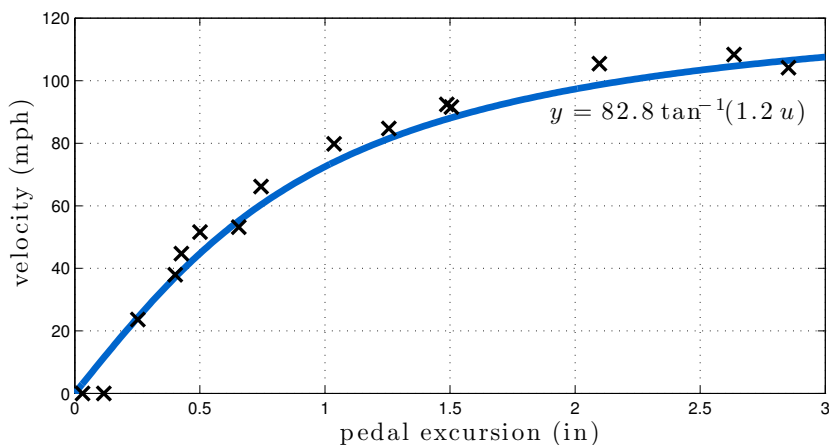


Fig. 1.3: Fitting the curve $y = \alpha \tan^{-1}(\beta u)$ to the data from Fig. 1.2

where u is the input, pedal excursion in inches, and y is the output, terminal velocity in mph. The parameters $\alpha = 82.2$ and $\beta = 1.2$ shown in Fig. 1.3 were obtained from a standard least-squares fit [LH87].

One of the most useful classes of mathematical models one can consider is that of *linear models*. One might be tempted to equate linear with simple. Whether this might be true in many cases, simplicity is far from a sin. More often than not, the loss of some feature neglected by a linear model is offset by the availability of a much broader set of analytic tools. It is often better to *know* when you are wrong than to *believe* you are right. This book is mostly concerned with linear models.

Speaking of linear models, one might propose to describe the experiment in Fig. 1.2 by a linear mathematical model of the form

$$y = \gamma u.$$

Fig. 1.4 show two of such models (dashed lines). The curve with coefficient $\gamma = 50.2$ was obtained by performing a least-square fit to all data points. The curve with coefficient $\gamma = 99.3$ is a first order approximation of the nonlinear model calculated in Fig. 1.3. Clearly, each model has its limitations in describing the experiment. Moreover, one model might be more suited to describe certain aspects of the experiment than the other. Responsibility rests with the user to make the best possible use of each model.

1.2 Cautionary Note

It goes without saying that the mathematical models described in § 1.1 do not purport to capture every single detail of the experiment, not to mention reality.

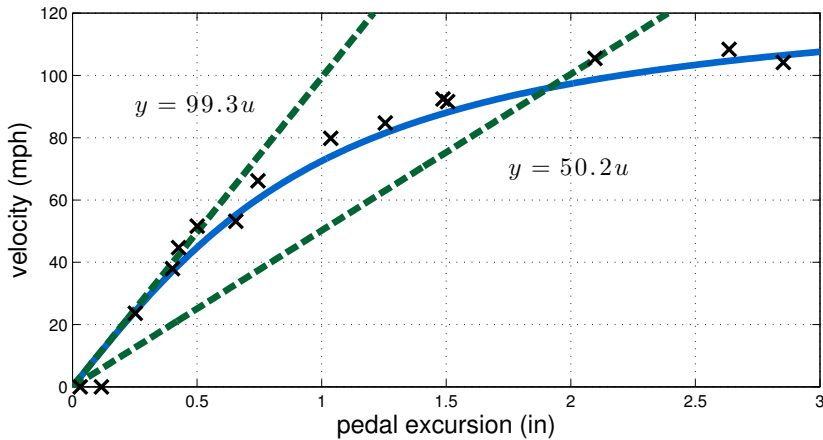


Fig. 1.4: Linear mathematical models of the form $y = \gamma u$ for the data in Fig. 1.2 (dashed); the model with $\gamma = 50.2$ was obtained by a least-squares fit; the model with $\gamma = 99.3$ was obtained after linearization of the nonlinear model (solid) obtained in Fig. 1.3

However, good models do capture essential aspects that we *perceive* as real, for example how the terminal velocity of a car responds to the acceleration pedal in the given experimental conditions. A model does not even need to be *correct* to be useful: for centuries men used² a model in which the Sun revolves around the Earth to predict and control the days! What is important is that models provide a way to express *some* aspects of reality using mathematics. When mathematical models are used in control design, it is therefore with the understanding that the model is bound to capture only a subset of relevant features of the actual phenomenon these models represent. At no time one should be fooled to *believe* in a model. The curious reader will appreciate [Fey86].

With this caveat in mind, it is often useful to think of an idealized *true* or *nominal model*, just as it is done in physics, against which a particular setup can be *mathematically* evaluated. This nominal model might be even different than the model used for computing a particular control algorithm, for instance, having more details or being more complex or accurate. Of course *physical* evaluation of a control system with respect to the underlying natural phenomenon is only possible by means of experimentation which should also include the physical realization of the controller in the form of computer hardware and software, electric circuits or other necessary mechanical devices.

²Apparently 1 in 4 Americans and 1 in 3 Europeans still go by that model [Gro14].

1.3 A Control Problem

Consider the following problem:

Under the experimental conditions described in § 1.1 and given a target terminal velocity, $\bar{y}$, is it possible to design a system, the controller, that is capable to command the accelerator pedal of a car, the input, u , to produce a terminal velocity, the output, y , equal to the target velocity?

An *automatic* system that can solve this problem is found in many modern cars with the name *cruise controller*. Of course, another *system* that is capable of solving the same problem is a human driver³. In this book we are mostly interested in solutions that can be implemented as an *automatic control*, that is, which can be performed by some combination of mechanical, electric, hydraulic or pneumatic systems running without human intervention, often being programmed in a digital computer or some other logical circuit or calculator.

Problems such as this are referred in the controls literature as *tracking* problems: the controller should make the system, a car, *follow* or *track* a given target output, the desired terminal velocity. In this section we will discuss two possible approaches to the cruise control problem.

1.3.1 Solution without Feedback

The role of the controller in tracking is to compute the input signal u which produces the desired output signal y . One might therefore attempt to solve a tracking problem using a system (controller) of the form:

$$u = K(\bar{y}).$$

The controller can only use the reference signal, the target output $\bar{y}$, and is said to be in *open-loop*⁴, as u is not directly a function of the output signal y .

With the intent of analysing the proposed solution using mathematical models, assume that the car can be represented by a *nominal model*, say G , that relates the input u (pedal excursion) to the output y (terminal velocity) through the mathematical function:

$$y = G(u).$$

The connection of the controller with this idealized model is depicted in the block diagram in Fig. 1.5. Here the function G can be obtained after fitting experimental data as done in Figs. 1.3 or 1.4, or borrowed from physics or engineering science principles. The block-diagram in Fig 1.5 represents the

³After some 16 or such years of *learning*.

⁴As opposed to *closed-loop* which will be discussed in § 1.3.2.

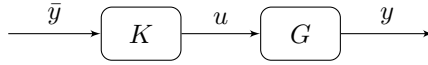


Fig. 1.5: Open-loop solution

following relationships:

$$y = G(u), \quad u = K(\bar{y}),$$

which can be combined to obtain

$$y = G(K(\bar{y})).$$

If G is *invertible* and K is chosen to be the inverse of G , that is $K = G^{-1}$, then

$$y = G(G^{-1}(\bar{y})) = \bar{y}.$$

Note that the match between the controller, K , and the nominal model, G , is paramount: if $K \neq G^{-1}$ then $y \neq \bar{y}$.

When both the nominal model G and the controller K are linear:

$$y = G u, \quad u = K \bar{y}, \quad y = G K \bar{y}.$$

from which $\bar{y} = y$ only if the product of the *constants* K and G is equal to one, that is, if

$$K G = 1 \quad \implies \quad K = \frac{1}{G}, \quad u = \frac{1}{G} \bar{y}.$$

Because the control law relies on knowledge of the nominal model G to achieve its goal, any imperfection in the model or in the implementation of the controller will lead to imperfect tracking.

1.3.2 Solution with Feedback

The open-loop solution considered in § 1.3.1 is only allowed to make use of the target output, $\bar{y}$. When a measurement, even if imprecise, of the system output, y , is available, one may allow the controller to make use of the measurement signal, y . In the case of the car cruise control, the terminal velocity, y , can be measured by an on-board odometer. Of course the target velocity, $\bar{y}$, is set by the driver.

Controllers that make use of output signals to compute the control inputs are called *feedback controllers*. In its most general form, a feedback controller has the functional form:

$$u = K(\bar{y}, y).$$

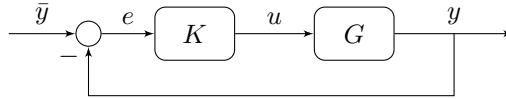


Fig. 1.6: Closed-loop feedback solution; circle represent the sum of signals with the indicated sign, i.e. $e = \bar{y} - y$

In practice, most feedback controllers make use of the signal y by first creating an *error signal*, $\bar{y} - y$, which is then used by the controller:

$$u = K(e), \quad e = \bar{y} - y. \quad (1.1)$$

This scheme is depicted in the block-diagram in Fig. 1.6. One should question whether it is possible to implement a system that replicates the block-diagram in Fig. 1.6. In the diagram, the measurement y that takes part in the computation of the control u in the controller block, K , is the same that comes out as a product of the system, G . In other words, the signals flow in this diagram is *instantaneous*. Even though we are not yet properly equipped to address such questions, we anticipate that it will be possible to construct and analyze *implementable* versions of the feedback diagram in Fig. 1.6 by taking into account dynamic phenomena, which we will start discussing in the next chapter.

At this point, we are content to say that if the computation implied by feedback is performed *fast enough*, then the scheme will work. We analyze the proposed feedback solution only in the case of linear models, that is, when both the controller, K , and the system to be controlled, G , are linear. Feedback controllers of the form (1.1), which are *linear* and *static*, are known by the name *proportional controllers*, or *P-controllers* for short. In the *closed-loop* diagram of Fig. 1.6, we can think of the signal $\bar{y}$, the target velocity, as an input, and of the signal y , the terminal velocity, as an output. A mathematical description of the relationship between the input signal, $\bar{y}$, and output signal, y , assuming linear models can be computed from the diagram:

$$y = G u, \quad u = K e, \quad e = \bar{y} - y$$

After eliminating the signals e and u we obtain

$$y = G K e = G K (\bar{y} - y) \quad \implies \quad (1 + G K) y = G K \bar{y}.$$

When $G K \neq -1$:

$$y = H \bar{y}, \quad H := \frac{G K}{1 + G K}.$$

The mathematical relationship governing a particular pair of inputs and outputs is called a *transfer-function*. The constant H above is called a *closed-loop transfer-function*.

		K				
		0.02	0.05	0.5	1	3
G	50.2	0.5010	0.7151	0.9617	0.9805	0.9934
	71.7	0.5892	0.7819	0.9729	0.9862	0.9954
	99.3	0.6651	0.8324	0.9803	0.9900	0.9967

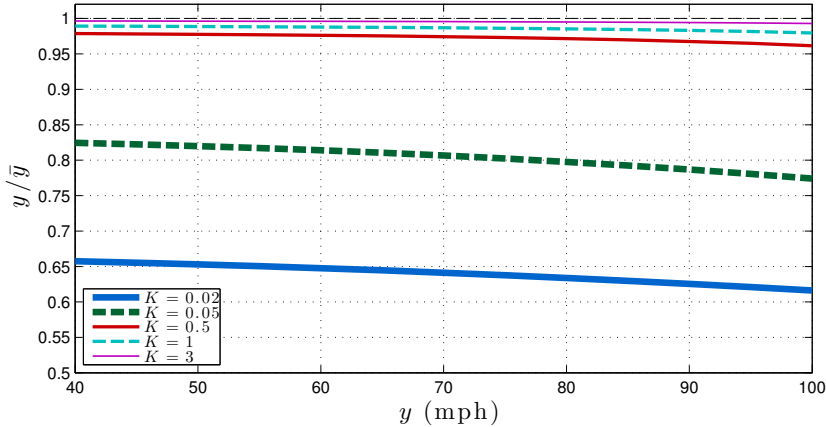
Tab. 1.1: Closed-loop transfer function for various values of K and G 

Fig. 1.7: Effect of the gain K in the ability of the terminal velocity, y , to track a given target velocity, $\bar{y}$, when the linear feedback control $u = K(y - \bar{y})$ is in closed-loop with the nonlinear model $y = G(u)$ as in Fig. 1.6, where $G(u) = 82.2 \tan^{-1}(1.2u)$ is from Fig. 1.3

Ironically, a first conclusion from the closed-loop analysis is that it is not possible to achieve exact tracking of the target velocity since H can not be equal to one for any finite value of the constants G and K . Not even when $K = G^{-1}$, which is the open-loop solution.

However, it is not so hard to make H get close to one: just make K large! More precisely, make the product GK large. How large it needs to be will depend on the particular system G . However, a welcome side-effect of the closed-loop solution is that the controller gain K does not depend directly on value of the model G . As the calculations in Tab. 1.1 reveal, the closed-loop transfer-function H remains within 1% of 1 for values $K > 3$ and for *any* value of G lying between the two linear crude linear models estimated earlier in Fig. 1.4.

One conclusion is that feedback control does not rely on knowledge of the exact model, G in this case, in order to achieve good tracking. This is a major feature of feedback control, and one of the reasons we shall often not need complete and sometimes not even accurate mathematical models for feedback de-

sign. This remarkable property remains true even if nonlinear⁵ models are considered, although the computation of H is more complicated⁶. Fig. 1.7 shows a plot of $y/\bar{y}$ for various choices of gain K when a linear controller is in feedback with the nonlinear model G fitted in Fig. 1.3. The result is virtually the same obtained using linear models.

Insight on the reasons why feedback control can achieve tracking without relying on precise models is obtained if we look at the control u that is effectively computed by the closed-loop. Following steps similar to the ones used in the derivation of the closed-loop transfer-function we calculate:

$$u = Ke = K(\bar{y} - y) = K(1 - H)\bar{y} = \frac{K}{1 + GK}\bar{y} = \frac{1}{K^{-1} + G}\bar{y}$$

Note that

$$\lim_{K \rightarrow \infty} u = G^{-1}\bar{y},$$

which is the exact same control computed in open-loop (see § 1.3.1). This time however, it is the feedback loop that *computes* G^{-1} based on the error signal $\bar{y} - y$. Indeed, u is simply equal to $K(\bar{y} - y)$ which will converge to $G^{-1}\bar{y}$ by virtue of feedback no matter the value of G when K is made large. A natural question is what are the side effects of raising the control gains in order to improve the tracking performance? We discuss this question in many points of this book as we learn more about dynamic systems and feedback.

1.3.3 Sensitivity

In the previous sections, we made a statement regarding how insensitive the closed-loop feedback solution was with respect to changes in the system model when compared to the open-loop solution. We can quantify this statement in the case of linear models.

As seen before, in both open- and closed-loop solutions to our tracking control problem the output y is related to the target output $\bar{y}$ through

$$y = H(G)\bar{y}.$$

The notation $H(G)$ indicates that the transfer-function H depends on the model G . In the open-loop solution $H(G) = GK$ and in the closed-loop solution $H(G) = GK(1 + GK)^{-1}$.

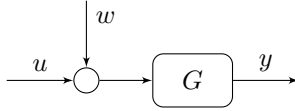
Now consider that G assumes values on a neighborhood of a certain nominal model $\bar{G}$ and that $H(\bar{G}) \neq 0$. Assume that those changes in G affect H in a continuous and differentiable way so that⁷

$$H(G) = H(\bar{G}) + H'(\bar{G})(G - \bar{G}) + O((G - \bar{G})^2)$$

⁵Many but not all nonlinear models.

⁶It requires solving the nonlinear algebraic equation $y = G(K(\bar{y} - y))$ for y .

⁷The notation $O(x^n)$ indicates a polynomial in x that has only terms with degree greater than or equal to n .

Fig. 1.8: System with input disturbance w

which is a Taylor Series expansion of H as a function of G . Discarding terms of order greater than or equal to two we can write

$$H(\bar{G}) - H(G) \approx H'(\bar{G})(\bar{G} - G)$$

After dividing by $H(\bar{G})$ we obtain an expression for how changes in G affect the transfer-function $H(G)$:

$$\frac{H(\bar{G}) - H(G)}{H(\bar{G})} \approx S(\bar{G}) \frac{\bar{G} - G}{\bar{G}}, \quad S(G) := \frac{G}{H(G)} H'(G).$$

The function S is called the *sensitivity* function.

Using this formula we compute the sensitivity of the open-loop solution in the case of linear models:

$$H(G) = GK, \quad \implies \quad S(G) = \frac{G}{GK} K = 1.$$

This can be interpreted as follows: in open-loop, a relative change in the system model, G , produces a relative change in the output, y , of the same order.

In closed-loop, after some calculations:

$$H(G) = \frac{GK}{1 + GK}, \quad \implies \quad S(G) = \frac{1}{1 + GK}.$$

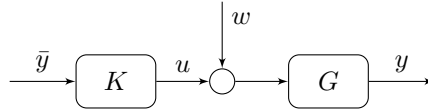
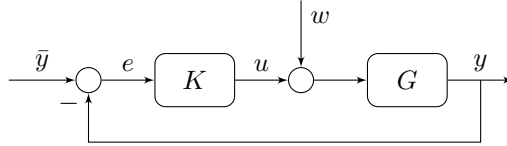
By making K large we not only improve tracking performance but also reduce the sensitivity S . Note that $S + H = 1$, hence $S = 1 - H$ so values of S can be obtained easily from Tab. 1.1 in the case of the car cruise control.

1.3.4 Disturbances

One form of accounting for variability in models is with the introduction of *disturbance signals*. Consider the block-diagram in Fig. 1.8 in which the disturbance signal w adds to the input signal u , that is:

$$y = G(u + w).$$

The disturbance signal w can be used to model signals that are present during the operation of the system, as well as phenomena acting on the system that are not fully modelled. For example, in § 2.7, we will use a disturbance signal to model a road slope in the car cruise control problem.

Fig. 1.9: Open-loop configuration with input disturbance w Fig. 1.10: Closed-loop feedback configuration with disturbance w

It is easy to incorporate disturbances to the basic open- and closed-loop schemes of Figs. 1.5 and 1.6, which we do in Figs. 1.9 and 1.10. In both cases, one can write the output y in terms of the reference input $\bar{y}$ and the disturbance w . Better yet, we can write the transfer-function from the inputs, $\bar{y}$ and w , to the tracking error, $e = \bar{y} - y$. In open-loop we calculate with Fig. 1.9:

$$e = \bar{y} - y = \bar{y} - G(K\bar{y} + w) = (1 - GK)\bar{y} - Gw$$

Substituting the proposed open-loop solution, $K = G^{-1}$, we obtain

$$e = -Gw$$

which means that open-loop control is very effective at tracking but has no capability to *reject* the disturbance w , as one could have anticipated from the block-diagram in Fig. 1.9. Open-loop controllers will perform poorly in the presence of disturbances. This is similar to the conclusion obtained in § 1.3.3 that showed open-loop controllers to be sensitive to changes in the system model.

In closed-loop, Fig. 1.10, we calculate:

$$e = \bar{y} - y = \bar{y} - G(K e + w) \quad \implies \quad (1 + GK)e = \bar{y} + Gw$$

and

$$e = \frac{1}{1 + GK} \bar{y} + \frac{G}{1 + GK} w. \quad (1.2)$$

In closed-loop, the control gain K shows up in both transfer-functions from the inputs, w and $\bar{y}$, to the tracking error, e . Furthermore, high control gain K reduces both terms at the same time. That is, the closed-loop solution achieves good tracking *and* rejects the disturbance signal w . This is a most welcome feature and often the main reason for using feedback in control systems. Coincidentally, the coefficient of the first term in (1.2) is the same as the sensitivity function, $S(G)$, calculated in § 1.3.3.

Problems

P1.1. For each block-diagram in Fig. 1.11 identify *inputs*, *outputs*, and other relevant *signals*, as well as describe what physical quantities the signals could represent. Tell whether the system is in *closed-loop* or *open-loop* based on the presence or absence of *feedback*. Is the relationship between the inputs and outputs dynamic or static? Write a simple equation for each block if possible. Which signals are *disturbances*?

P1.2. Sketch block-diagrams that can represent the following phenomena as *systems*:

- a) skin protection from sunscreen;
- b) money on a savings account;
- c) a chemical reaction;
- d) bitterness of your morning coffee;

Identify potential *input* and *output* signals that can be used to identify a cause-effect relationship. Discuss the assumptions and limitations you used in the construction of your *model*. Is the relationship between the inputs and outputs dynamic or static? Write simple equations for each block if possible.

P1.3. Mammals are able to regulate their body temperature near 36.5°C ($\sim 98^{\circ}\text{F}$) despite large fluctuations in the ambient temperature. Sketch a block-diagram that could represent a possible temperature control system in mammals. Identify disturbances, signals and possible principles of sensing and actuation that could be used by mammals to lower or increase the body temperature. Compare possible *open-loop* and *closed-loop* solutions. Discuss the difficulties that need to be overcome in each case.

P1.4. Most cars nowadays are equipped with an *anti-lock breaking system* (ABS), which are designed to prevent the wheels from *locking up* when the driver actuates the break pedal. It helps with emergency breaking and breaking under adverse road conditions by ensuring that traction is maintained on all wheels throughout breaking. An ABS system detects potential locking of a wheel by comparing the rotational speeds among wheels and increases or reduces the pressure on that particular wheel hydraulic break actuator as needed. Sketch a block-diagram that could represent the signals and systems involved in ABS.

P1.5. Humans learn to walk early in life. Sketch a block-diagram that could represent signals and systems required for navigate accros an unknown

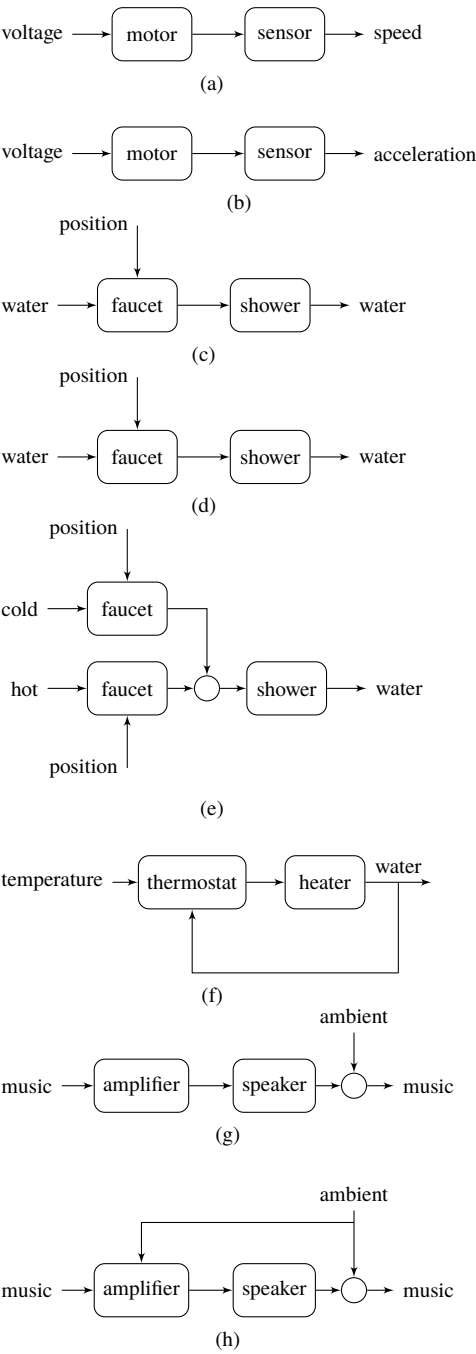
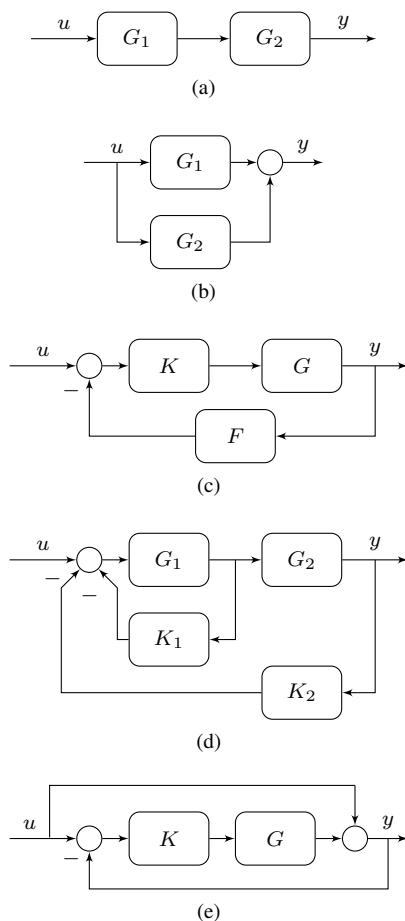


Fig. 1.11: Block diagrams for P1.1

Fig. 1.12: Block diagrams for **P1.7**

environment. What sensors are used? Is there feedback involved?

P1.6. Repeat **P1.5** this time from the perspective of a blind person.

P1.7. For each block-diagram in Fig. 1.12 compute the transfer function from the input u to the output y assuming that the blocks are linear.

P1.8. Students taking part in the Galileo Project at Rice University set out to carefully reproduce some of Galileo's classic experiments [Jen14]. One was the study of projectile motion using an inclined plane, in which a ball accelerates down a plane inclined at a certain angle then rolls in the horizontal direction with uniform motion for

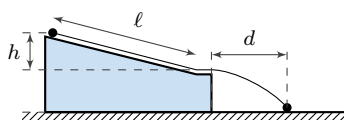


Fig. 1.13: Galileo's inclined plan

#	Ramp distance @ 13.4°			
	1 ft	2 ft	4 ft	6 ft
1	$13\frac{15}{16}$	$19\frac{13}{16}$	$27\frac{11}{16}$	$33\frac{3}{8}$
2	$13\frac{7}{8}$	$19\frac{13}{16}$	$27\frac{3}{4}$	$33\frac{5}{16}$
3	$14\frac{1}{16}$	$19\frac{13}{16}$	$27\frac{3}{4}$	$33\frac{3}{16}$
4	14	$19\frac{3}{4}$	$27\frac{9}{16}$	$33\frac{7}{16}$
5	$13\frac{15}{16}$	$19\frac{3}{4}$	$27\frac{9}{16}$	$33\frac{5}{8}$

#	Ramp distance @ 6.7°				
	1 ft	2 ft	4 ft	6 ft	8 ft
1	$10\frac{11}{6}$	$14\frac{1}{2}$	$20\frac{3}{4}$	$25\frac{7}{16}$	$29\frac{5}{8}$
2	$10\frac{11}{6}$	$14\frac{9}{16}$	$20\frac{3}{4}$	$25\frac{1}{2}$	$29\frac{1}{2}$
3	$10\frac{11}{6}$	$14\frac{1}{2}$	$20\frac{3}{4}$	$25\frac{3}{4}$	$29\frac{1}{2}$
4	$10\frac{11}{6}$	$14\frac{1}{2}$	$20\frac{3}{4}$	$25\frac{1}{2}$	$29\frac{5}{16}$
5	$10\frac{11}{6}$	$14\frac{9}{16}$	$20\frac{3}{16}$	$25\frac{5}{8}$	$29\frac{1}{2}$

Tab. 1.2: Data for **P1.8**

a short while until falling off the edge of a table, as shown in Fig. 1.13. The distance the ball rolled along the inclined plane, ℓ in feet, and the distance from the end of the table to the landing site of the ball, d in inches, were recorded. Some of their data, 5 trials at two different angles, is reproduced in Tab. 1.2. Use `matlab` to plot and visualize the data. Fit simple equations, e.g. linear, quadratic, etc., to the data to relate the fall height, h , to the horizontal travel distance, d , given in Tab. 1.2. Justify your choice of equations and comment on the quality of the fit obtained in each case. What other information can you obtain from this experiment knowing that the free fall was always initiated 40 inches above the ground?

Chapter 2

Dynamic Systems

In § 1 we contemplated solutions to our first control problem, a much simplified cruise controller for a car, without taking into account possible effects of *time*. System and controller models were *static* relations between the signals: the output signal y , the input signal u and the reference input $\bar{y}$. Signals in block-diagrams flow *instantaneously* and closed-loop solutions derived from such block-diagrams were deemed reasonable only if they could be implemented *fast enough*. We drew encouraging conclusions from simple analysis but no reason was given to support these conclusions if time is taken into consideration.

Of course it is perfectly fine to construct a static mathematical model relating a car's pedal excursion with its terminal velocity, as long as we understand that this phenomenon is *not* instantaneous. Clearly a car cannot accelerate and reach a terminal velocity *instantaneously*! If we expect to implement the feedback cruise controller in a real car, we have to be prepared to say what happens between the time a terminal target velocity is set and the time the car reaches its terminal velocity. In many cases controllers will have to *understand* that the car does not reach its terminal velocity instantaneously. That is, we will have to incorporate time not only to models and tools but also to controllers. For this reason we need to learn how to work with *dynamic systems*.

In the present book, mathematical models for dynamic systems will take the form of ordinary differential equations where signals evolve continuously in time. Bear in mind that this is not a course on differential equations, and previous exposure to the mathematical theory of differential equations is helpful. Familiarity with material covered in standard text books, e.g. [BD12], is enough for understanding the concepts to be addressed here. We make extensive use of the Laplace transform and provide a somewhat self-contained review of relevant facts in § 3.

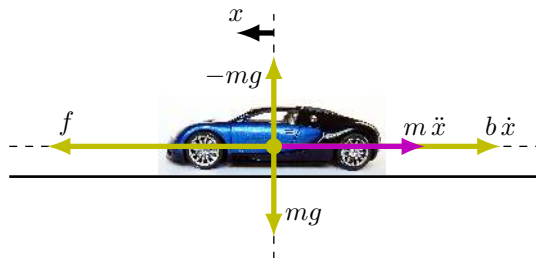


Fig. 2.1: Free-body diagram showing forces acting in a car

2.1 Dynamic Models

Let us start with some notation: we denote time by the real variable t in the interval $[0, \infty)$, where 0 can be thought as an arbitrary origin of time before which we are not interested in the behavior of the system nor its signals. We employ functions of real variables to describe signals and use standard functional notation to indicate the dependency of signals on time. For example, the dynamic signals y and u are denoted as $v(t)$ and $u(t)$. At times, when no confusion is possible, we omit the dependence of signals on t in order to lighten up the notation.

We claimed in § 1 that models should be rooted in well defined experiments. Planning and performing experiments for dynamic systems is a much more complex task, which we do not have room to address in detail here. Instead, we will reach out to physics to help us introduce an abstract dynamic model, the parameters of which will be later determined through measurement and experimentation.

In the tradition of simplified physical modelling, we use Newton's Law to write equations for a car based on the free-body diagram shown in Fig. 2.1. The car is modeled as a particle with mass $m > 0$ and after balancing all forces in the x -direction we obtain the differential equation:

$$m\ddot{x}(t) + b\dot{x}(t) = f(t),$$

where x is the linear coordinate representing the position of the car, $b \geq 0$ is the coefficient of friction, and f is a force, which we will use to put the the car in movement. Much can be argued about the exact form of the friction force, which we have assumed to be *viscous*, that is of the form $-bv(t)$, linear and opposed to the velocity $v(t) = \dot{x}(t)$. As we are interested in modeling the velocity of the car and not its position, it is convenient to rewrite the differential equation in terms of the velocity v , obtaining:

$$m\dot{v}(t) + bv(t) = f(t). \quad (2.1)$$

In order to complete our model we need to relate the car driving force, f , to the pedal excursion, u . Here we could resort to experimentation or appeal to basic

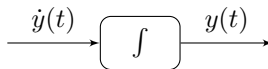


Fig. 2.2: Block-diagram with integrator

principles. With simplicity in mind we choose a linear static model:

$$f = p u \quad (2.2)$$

where p represents a *pedal gain*, which can be determined experimentally by methods similar to the ones used in § 1.

Of course, no one should believe that the simple force model (2.2) can accurately represent the response of the entire powertrain of the car in a variety of conditions. Among other things, the powertrain will have its own complex dynamic behaviors, which (2.2) gracefully ignores. Luckily, the validity¹ of such simplification depends not only on the behavior of the actual powertrain but also on the purpose of the model. In many cases, the time-constants² of the powertrain are much *faster* than the time-constant due to the inertial effects of the entire car. In this context, a simplified model can lead to satisfactory or at least insightful results when the purpose of the model is, say, predicting the velocity of the car. A human driver certainly does not need to have a deep knowledge of the mechanical behaviour of an automobile for driving one!

Combining equations (2.1) and (2.2), and labeling the velocity as the output of the system, i.e. $y(t) = v(t)$, we obtain the differential equation:

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t). \quad (2.3)$$

which is the dynamic model we will use to represent the car in the dynamic analysis of the cruise control problem.

2.2 Block-Diagrams for Differential Equations

In § 1 we used block-diagrams to represent the interaction of signals, systems, and controllers. If we interpret differential equations as a relationship between signals and the signals' derivatives, it should be no surprise that ordinary differential equations can be represented in block-diagrams. The key is to use an *integrator* to relate the derivative to its primitive. An integrator block should be able to produce at its output the integral of its input. Alternatively, we can see the integrator input as the derivative of its output signal, as depicted in Fig. 2.2. We will study in § 5 how a number of physical devices can be used to implement integrators.

¹A better word here may be *usefulness*.

²More about that soon!

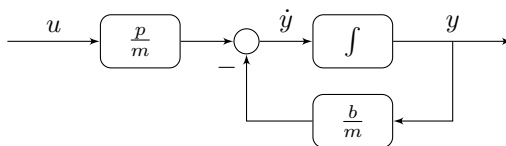


Fig. 2.3: Dynamic model of the car: m is the mass, b is the viscous friction coefficient, p is the pedal gain, u is the pedal excursion, y is the car's velocity

Assuming that integrator blocks are available, all that is left to do is to rewrite the ordinary differential equation isolating its highest derivative. For example, we rewrite (2.3) as:

$$\dot{y}(t) = \frac{p}{m} f(t) - \frac{b}{m} y(t),$$

which can be represented by the block-diagram in Fig. 2.3.

Note the presence of a *feedback-loop* in the diagram of Fig. 2.3! For this reason, tools for analyzing feedback loops many times draw on the theory of differential equations and vice-versa. We will explore the realization of differential equations using block-diagrams with integrators in detail in § 5.

2.3 Dynamic Response

The differential equation (2.3) looks very different from the static models considered earlier. In order to understand their differences and similarities we need to understand how the model (2.3) *responds* to inputs. Our experiment in § 1 consisted of having a constant pedal excursion and letting the car reach a terminal velocity. We shall first attempt to emulate this setup using the differential equation (2.3) as a model.

Applying a constant pedal excursion corresponds to the input function:

$$u(t) = \tilde{u}, \quad t \geq 0$$

where $\tilde{u}$ is constant. In response to this input we expect that the solution of the differential equation (2.3) approaches a constant terminal velocity, $\tilde{y}$, after some time has passed. The value of this terminal velocity can be calculated after noticing that

$$y(t) = \tilde{y} \quad \implies \quad \dot{y}(t) = 0$$

in which case (2.3) reduces to

$$\frac{b}{m} \tilde{y} = \frac{p}{m} \tilde{u}.$$

In other words

$$\tilde{y} = \frac{p}{b} \tilde{u}.$$

It is this relation that should be compared with the static model developed earlier. Experiments similar to the ones in § 1.1 can be used to determine an adequate value for the ratio p/b .

In the language of differential equations, the function

$$y_P(t) = \tilde{y} = \frac{p}{b} \tilde{u}, \quad t \geq 0$$

is a *particular solution* to the differential equation (2.3). This particular solution cannot however be complete: if the initial velocity of the car at time $t = 0$ is not equal to $\tilde{y}$ then $y(0) = y_0 \neq \tilde{y} = y_P(0)$. The remaining component of the solution is found³ by solving the homogeneous version of equation (2.3):

$$\dot{y}(t) + \frac{b}{m} y(t) = 0.$$

All solutions to this equation are of the form

$$y_H(t) = e^{\lambda t}$$

where the constant λ is determined upon substitution of $y_H(t)$ into the differential equation, producing

$$\dot{y}_H(t) + \frac{b}{m} y_H(t) = \left(\lambda + \frac{b}{m} \right) e^{\lambda t} = 0.$$

This equation needs to hold for all t , in particular $t = 0$. This will happen only if λ is a zero of the *characteristic equation*:

$$\lambda + \frac{b}{m} = 0 \quad \implies \quad \lambda = -\frac{b}{m}. \quad (2.4)$$

The complete solution to equation (2.3) is a combination of the particular solution, $y_P(t)$, with all possible solutions to the homogeneous equation (in this example there is only one, $y_H(t)$), that is:

$$y(t) = y_P(t) + \beta y_H(t).$$

The constant β is calculated so that $y(t)$ matches the *initial condition*, y_0 , at $t = 0$. That is:

$$y(0) = \tilde{y} + \beta = y_0 \quad \implies \quad \beta = y_0 - \tilde{y}.$$

The complete response takes the form:

$$y(t) = \tilde{y} (1 - e^{\lambda t}) + y_0 e^{\lambda t}, \quad t \geq 0, \quad (2.5)$$

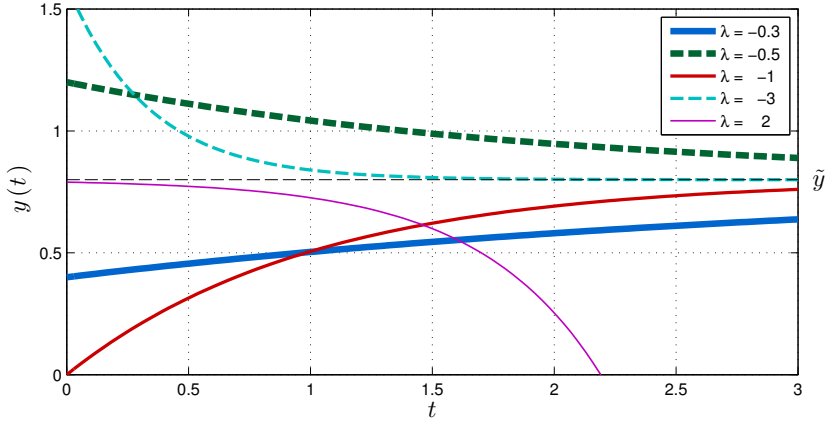


Fig. 2.4: Plots of $y(t) = \tilde{y}(1 - e^{\lambda t}) + y_0 e^{\lambda t}$, for $t \geq 0$ with $\tilde{y} = 0.8$, and various values of λ and y_0

where

$$\lambda = -\frac{b}{m}, \quad \tilde{y} = \frac{p}{b}\tilde{u}. \quad (2.6)$$

Plots of $y(t)$ for various values of λ , $\tilde{y}$, and y_0 are shown in Fig. 2.4. Note how the responses *converge* to $\tilde{y}$ for all negative values of λ and that the more negative the value of λ the faster the convergence. When λ is positive the response does not converge to $\tilde{y}$, even when y_0 is very close to $\tilde{y}$.

It is customary to evaluate how fast the solution of the differential equation (2.3) converges to $\tilde{y}$ by analysing its response to a zero initial condition $y_0 = 0$ and a nonzero $\tilde{y} \neq 0$. For reasons that will be clear later in § 3, this is known as a *step response*. When $\lambda > 0$, the constant

$$\tau := -\lambda^{-1}.$$

is the *time-constant*. At $t = \tau$

$$y(\tau) = \tilde{y}(1 - e^{-1}) \approx 0.63 \tilde{y}.$$

At $t = 3\tau$

$$y(3\tau) = \tilde{y}(1 - e^{-3}) \approx 0.95 \tilde{y}.$$

As τ depends only on λ , it is possible to compare the rate of convergence of different systems modeled by differential equations by comparing their time-constants. For differential equations more complex than (2.3), the time-constant

³See [BD12] for details.

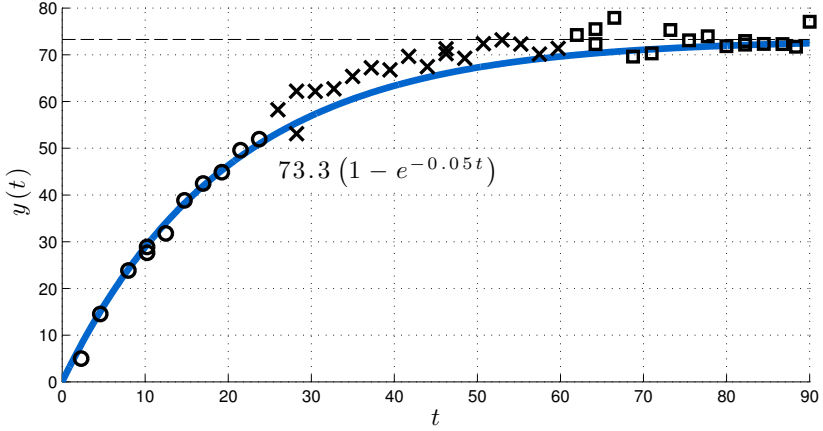


Fig. 2.5: Experimental velocity response of a car to a constant pedal excursion, $u(t) = \tilde{u} = 1$ in, $t \geq 0$; samples are marked as circles, crosses and squares

is *defined* as the time it takes the step response to reach 63% of its terminal value. The smaller the time-constant, the faster the convergence.

Another measure of the speed of the response is the *rise-time*, t_r , which is the time it takes the step response to go from 10% to 90% of its final *steady-state* value. That is

$$t_r = t_2 - t_1,$$

where

$$y(t_1) = \tilde{y}(1 - e^{\lambda t_1}) = 0.1 \tilde{y}, \quad y(t_2) = \tilde{y}(1 - e^{\lambda t_2}) = 0.9 \tilde{y},$$

from which

$$t_r = t_2 - t_1 = \ln(1/9) \lambda^{-1} = \ln(9) \tau \approx 2.2 \tau. \quad (2.7)$$

Again, the smaller the rise-time, the faster the convergence to steady-state.

2.4 Experimental Dynamic Response

Recall the experiment proposed in § 1.1 to estimate a static model for the car. The terminal velocity in the dynamic model is related to the static model parameter through $\tilde{y} = p/b$, which can be determined in the same way as done in § 1.1. A new experiment is needed to determine the parameter $\lambda = -b/m$, which does not influence the terminal velocity but affects the rate at which the car approaches the terminal velocity.

First let us select a velocity around which we would like to *build* our model. This should be a velocity close to the expected operation of the cruise controller.

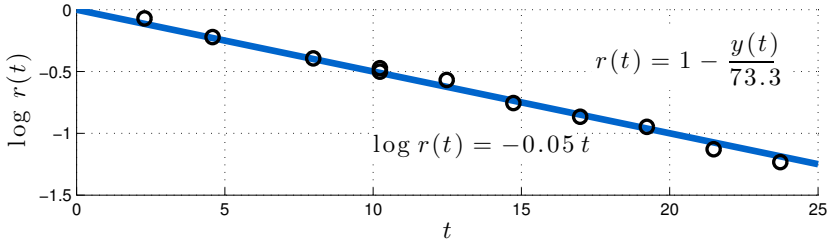


Fig. 2.6: Plot of $\log r(t) = \log[1 - y(t)/\tilde{y}] = \lambda t$ and fitted linear model

Looking at Figs 1.2 through 1.4, we observe that a pedal excursion of around 1 in will lead to a terminal velocity close to 70mph, which is close to highway speeds in which a cruise controller is expected to perform. We perform the following dynamic experiment: starting at rest, apply a constant pedal excursion

$$u(t) = \tilde{u} = 1 \text{ in}, \quad t \geq 0$$

and collect samples of the instantaneous velocity until the velocity becomes approximately constant. In other words, perform an experimental *step response*. The result of one such experiment looks like the plot in Fig 2.5, in which samples (circles, crosses, and squares) have been collected approximately every 2s for 90s.

We proceed by *fitting* the data in Fig. 2.5 to a function of the type (2.5) where y_0 is set to 0 to estimate the parameters $\tilde{y}$ and λ . This can be done in many ways. We do it as follows: we first average the samples over the last 30s in Fig 2.5 (squares) to compute an estimate of the terminal velocity. From the data shown in Fig 2.5 we obtain the estimate

$$\tilde{y} \approx 73.3 \text{ mph}.$$

If $y(t)$ is of the form (2.5) then

$$r(t) = 1 - y(t)/\tilde{y} = e^{\lambda t} \quad \implies \quad \log r(t) = \lambda t.$$

That is, $\log r(t)$ is a line with slope λ from which we estimate

$$\lambda = -0.05$$

after fitting a line to $\log r(t)$ in Fig. 2.6 computed using samples taken from the first 25s from Fig. 2.5 (circles). The parameters b/m and p/m are then estimated based on the relationships:

$$\begin{aligned} \frac{b}{m} &= -\lambda \approx 0.05 \text{ s}^{-1}, & \frac{p}{b} &= \frac{\tilde{y}}{\tilde{u}} \approx 73.3 \text{ mph/in}, \\ \frac{p}{m} &= \frac{b}{m} \times \frac{p}{b} = -\lambda \times \frac{\tilde{y}}{\tilde{u}} = 3.7 \text{ mph}/(\text{in s}) \end{aligned} \quad (2.8)$$

Note that this model has a static gain of about 73.3mph/in which lies somewhere between the two linear models estimated earlier in Fig. 1.4.

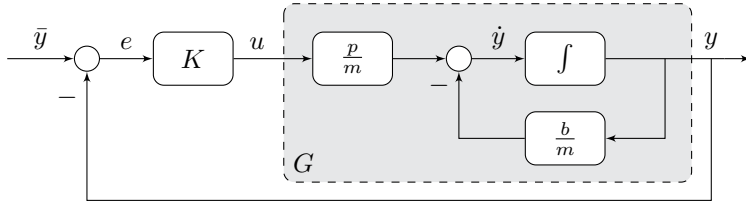


Fig. 2.7: Dynamic closed-loop connection of the car model with proportional controller

2.5 Dynamic Feedback Control

We are now ready to revisit the feedback solution proposed in § 1.3 for solving the cruise control problem. Let us keep the structure of the feedback loop the same, that is let the controller

$$u(t) = K e(t), \quad e(t) = \bar{y} - y(t) \quad (2.9)$$

be connected as in Fig. 1.6. Note a fundamental difference between this controller and the one analyzed before: in § 1.3.2 the signals e and y were the terminal error and terminal velocity; controller (2.9) uses the *dynamic* error signal $e(t)$ and velocity $y(t)$. This dynamic feedback loop can be practically implemented if a sensor for the instantaneous velocity, $y(t)$, is used. Every vehicle comes equipped with one such sensor, the odometer⁴. A dynamic linear controller of the form (2.9) is called a *proportional controller*.

In order to analyze the resulting dynamic feedback control loop we replace the system model G with the dynamic model (2.3) to account for the car's dynamic response to changes in the pedal excursion. In terms of block-diagrams, Fig. 2.3 replaces G in Fig. 1.6. The resulting block-diagram is shown in Fig. 2.7. Using equations (2.9) and (2.3) we eliminate the signal $u(t)$ to produce

$$\dot{y}(t) + \frac{b}{m}y(t) = \frac{p}{m}K(\bar{y} - y(t))$$

from which

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m}K \right) y(t) = \frac{p}{m}K\bar{y}. \quad (2.10)$$

This ordinary differential equation governs the behaviour of the closed-loop system. In the next chapters, we will see how it is possible to interpret this equation in terms of a closed-loop transfer-function, as done in the static case.

⁴The odometer measures the speed but it is easy to infer the direction, hence the velocity, in this simple one-dimensional setup.

K	$H(0)$	$S(0)$	$\tilde{y}$ (mph)	τ (s)	t_r (s)
open-loop	1.00	1.00	60	20.0	44.0
0.02	0.60	0.40	36	8.1	17.7
0.05	0.79	0.21	47	4.3	9.3
0.50	0.97	0.03	58	0.5	1.2

Tab. 2.1: Open- and closed-loop steady-state transfer-function, $H(0)$, steady-state sensitivity, $S(0)$, steady-state limit, $\tilde{y}$, time-constant, τ , and rise time, t_r , calculated for $b/m = 0.05$ and $p/b = 73.3$ and a constant target output of $\bar{y} = 60$ mph. The open-loop solution is from § 1.3.1

Since equation (2.10) has the same structure as equation (2.3), its solution is also given by (2.5). That is

$$y(t) = \tilde{y} (1 - e^{\lambda t}) + y_0 e^{\lambda t}, \quad t \geq 0$$

$$\lambda = -\frac{b}{m} - \frac{p}{m}K, \quad \tilde{y} = \frac{\frac{p}{m}K}{\frac{b}{m} + \frac{p}{m}K} \bar{y} = \frac{\frac{p}{b}K}{1 + \frac{p}{b}K} \bar{y}.$$

When K is positive, the decay-rate λ is negative and hence $y(t)$ converges to $\tilde{y}$. In terms of the target velocity $\bar{y}$ we can write:

$$\lim_{t \rightarrow \infty} y(t) = \tilde{y} = H(0) \bar{y}, \quad H(0) = \frac{\frac{p}{b}K}{1 + \frac{p}{b}K}. \quad (2.11)$$

We refer to components of the response of a dynamic system that persist as t grows large as *steady-state solutions*. In this case, the closed-loop has a constant steady-state solution

$$y_{ss}(t) = \tilde{y} = H(0) \bar{y},$$

where the value of $H(0)$ coincides with the static closed-loop transfer-function computed in § 1.3.2 if G replaces p/b . In terms of *tracking error*:

$$\lim_{t \rightarrow \infty} e(t) = \lim_{t \rightarrow \infty} \bar{y} - y(t) = \bar{y} - \tilde{y} = S(0) \bar{y}, \quad (2.12)$$

where

$$S(0) = 1 - H(0) = \frac{1}{1 + \frac{p}{b}K}, \quad (2.13)$$

and $S(0)$ is the closed-loop steady-state *sensitivity function*. The reason for the notation $H(0)$ and $S(0)$ will become clear in later chapters.

For various values of $G = p/b$, including $p/b \approx 73.3$ which we estimated in § 2.4, the steady-state closed-loop solution will track the reference $\bar{y}$ with accuracy $S(0) = 1 - H(0)$, which can be computed from the values listed

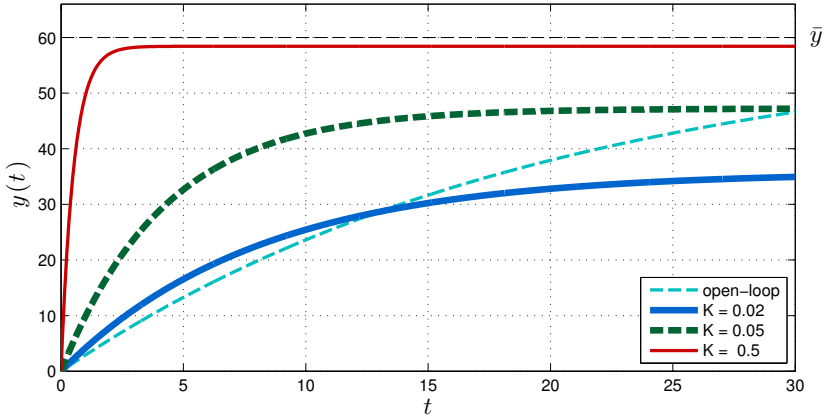


Fig. 2.8: Open- and closed-loop dynamic responses, $y(t)$, for the linear car velocity model (2.10) calculated for $b/m = 0.05$ and $p/b = 73.3$ and a constant target output of $\bar{y} = 60$ mph with proportional control. The open-loop solution is from § 1.3.1

in Tab. 1.1. In steady-state, the closed-loop behaves as predicted by the static analysis in § 1.3. The dynamic analysis goes a step further: it predicts the rate at which convergence occurs, which is related to

$$\lambda = -\frac{b}{m} - \frac{p}{m}K.$$

The time-constant $\tau = -\lambda^{-1}$ becomes smaller as K grows. A numerical comparison for various values of K and the open-loop solution (§ 1.3.1) is given in Tab. 2.1. The corresponding dynamic responses calculated from zero initial conditions, $y(0) = 0$, are plotted in Fig. 2.8.

Some numbers in Tab. 2.1 and Fig. 2.8 look suspicious. Is it really possible to lower time-constants so much? Take for example the case $K = 0.5$: you have almost perfect tracking (3% error) more than 40 times faster than open-loop. This kind of performance improvement is unlikely to be achieved by any controller that simply steps into the accelerator pedal. Surely there must be a catch! Indeed, so far we have been looking at the system output, the car's velocity, $y(t)$, and paid little attention to the control input, the pedal excursion, $u(t)$. We now look at the control input in search of clues for the impressive performance of the closed-loop controller.

The control inputs associated with the dynamic responses in Fig. 2.8 are plotted in Fig. 2.9. In Fig. 2.9 we see that the feedback controller is injecting into the system, the car, large inputs, pedal excursions, in order to achieve better tracking and faster response. The larger the control gain, K , the larger the required pedal excursion. Note that in this simple case the maximum required

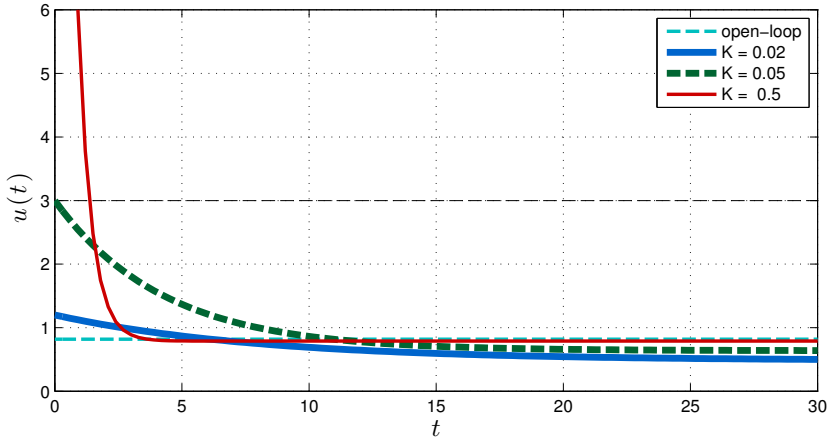


Fig. 2.9: Open- and closed-loop control inputs (pedal excursion) that produced the car linear dynamic responses in Fig. 2.8; the largest possible pedal excursion is 3 in

control signal happens at $t = 0$, when the tracking error is at a maximum and

$$u(0) = Ke(0) = K(\bar{y} - y(0)) = K\bar{y}.$$

Clearly, the larger K the larger will be the required control input. For instance, at $K = 0.5$ the controller produces an input that exceeds the maximum possible pedal excursions of 3 in, which corresponds to *full throttle*. With $K = 0.05$ we have $u(0) = 3$, which is at the limit of saturation. Of course, any conclusions drawn at this point for $K > 0.05$ will no longer be valid or, at least, not very accurate. It is not possible to achieve some of the predicted ultra fast response times due to limitations in the control system, in this case pedal and engine saturation, that were not represented in the linear models used to design and analyze the closed-loop. Ironically, the gain $K = 0.02$ is one in which the pedal excursion remains well below saturation, and is perhaps the one case in which the (poor) performance predicted by the linear model is likely to be accurate.

2.6 Nonlinear Models

In § 2.5 we saw controllers that produced inputs that led to saturation of the system input, the car's pedal excursion. In some cases the required control input exceeded full throttle. In this section we digress a little to introduce a simple nonlinear model that can better predict the behavior of the system in closed-loop when saturation is present. This is not a course in nonlinear control, and the discussion will be kept at a very basic level. The goal is to be able to tell

what happens in our simple example when the control input and the system reaches saturation.

In order to model the effect of saturation we will work with a nonlinear differential equation of the form:

$$\dot{y}(t) + \frac{c}{m} \tan(\alpha^{-1}y(t)) = \frac{d}{m}u(t), \quad u(t) \in [0, 3]. \quad (2.14)$$

When $u(t)$ is constant, i.e. $u(t) = \tilde{u} \in [0, 3]$, one particular solution to (2.14) is $y(t) = \tilde{y}$, where

$$\frac{c}{m} \tan(\alpha^{-1}\tilde{y}) = \frac{d}{m}\tilde{u} \quad \implies \quad \tilde{y} = \alpha \tan^{-1}(\beta\tilde{u}), \quad \beta = \frac{d}{c}.$$

This means that the steady-state response of the nonlinear differential equation (2.14) matches the empirical nonlinear fit that we performed earlier for the car model in Fig. 1.3. Moreover, at least for small values of $y(t)$, we should expect⁵ that

$$c \tan(\alpha^{-1}y(t)) \approx b y(t), \quad b = \alpha^{-1}c$$

Intuitively, as long as y remains small and $u(t) \in [0, 3]$, the dynamic response of the nonlinear differential equation (2.14) should stay *close* to the dynamic response of the linear differential equation (2.3).

In order to estimate suitable parameters c/m , d/m and α , we proceed as follows: first we borrow α and β from our previously computed nonlinear static fit ($\alpha = 82.8$ mph/in, $\beta = 1.2$ in⁻¹, see Fig. 1.3), then estimate b/m from the linear dynamic experiment described in § 2.4, and use the following relationships to estimate the nonlinear parameters:

$$\begin{aligned} \frac{c}{m} &= \alpha \times \frac{b}{m} = 82.8 \times 0.05 = 4.1 \text{ mph/in}, \\ \frac{d}{m} &= \beta \times \frac{c}{m} = 1.2 \times 4.1 = 5.0 \text{ mph/(in s)}. \end{aligned} \quad (2.15)$$

The rationale for the above *ad-hoc* fit is so that the nonlinear model matches the static fit, parameters α and β in Fig. 1.3, as well as the time-constant of the linear model, parameter b/m .

It is in closed-loop however that the nonlinear model will likely exposes serious limitations of our previously developed control algorithms based on the linear model (2.3). In order to better capture the limits on pedal excursion we introduce the additional explicit saturation nonlinearity:

$$\text{sat}_{(\underline{u}, \overline{u})}(u(t)) = \begin{cases} \overline{u}, & u(t) \geq \overline{u} \\ u(t), & \underline{u} \leq u(t) < \overline{u} \\ \underline{u}, & u(t) < \underline{u} \end{cases}$$

⁵This notion will be formalized later in § 5.

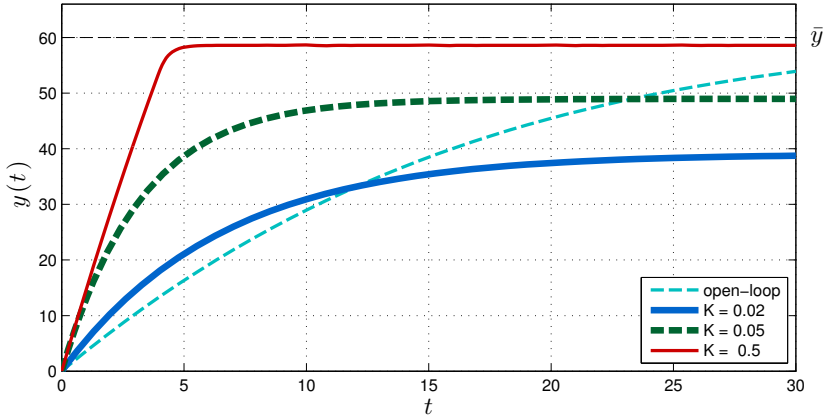


Fig. 2.10: Open- and closed-loop dynamic responses $y(t)$ produced by the nonlinear car velocity model (2.16) (solid) calculated with $\alpha = 1.2$, $\beta = 82.8$, $c/m = 4.1$ and $d/m = 5.0$ and a constant target output of $\bar{y} = 60$ mph with proportional control and various values of K . Compare with Fig. 2.8

In our model $\underline{u} = 0$ and $\bar{u} = 3$ in. The complete nonlinear model we use to simulate the system in closed-loop is of the form:

$$\dot{y}(t) + \frac{c}{m} \tan(\alpha^{-1}y(t)) = \frac{d}{m} \text{sat}_{(0,3)}(u(t)), \quad u(t) = K(\bar{y} - y(t)).$$

Eliminating $u(t)$ we obtain

$$\dot{y}(t) + \frac{c}{m} \tan(\alpha^{-1}y(t)) = \frac{d}{m} \text{sat}_{(0,3)}(K(\bar{y} - y(t))). \quad (2.16)$$

The above nonlinear ordinary differential equation as well as the open-loop equation (2.14) can be simulated using standard numerical integration methods, e.g. a third order Runge-Kutta method [BD12].

In order to see the effect of the nonlinearities in the closed-loop performance we repeat the simulations performed in Figs. 2.8 and 2.9, this time using the nonlinear feedback model (2.16). We show in Figs. 2.10 and 2.11 the nonlinear system closed-loop responses for various values of gains. These compare with Figs. 2.8 and 2.9. For values of $K = 0.05$ and $K = 0.02$, when the control input predicted using the linear model is within the linear region, i.e. $u(t) \in [0, 3]$, and the speeds are small, the nonlinearity has a minor impact. However, in the case of the larger gain $K = 0.5$, where the control input is heavily saturated (see Fig. 2.11), the response is significantly different. In particular, the extraordinarily fast response predicted by the linear model is not realized in the nonlinear model. In this simple example, the slower response seems to be the only apparent consequence. This will not always be the case, and severe nonlinearities will often negatively affect stability as well as performance of closed-loop systems.

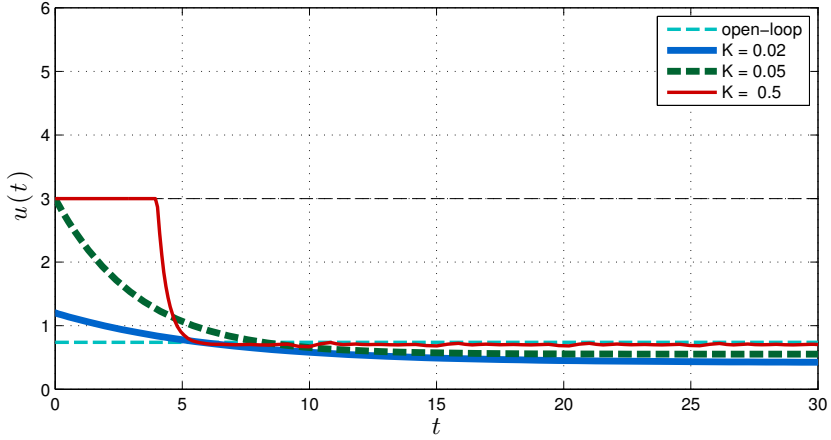


Fig. 2.11: Open- and closed-loop control inputs, pedal excursion, $u(t)$, produced by the car velocity nonlinear model eq. (2.16); the largest possible pedal excursion is 3 in; note the marked effect of pedal saturation in the case of the highest gain $K = 0.5$ and its impact on the response in Fig. 2.10

2.7 Disturbance Rejection

We now return to linear models to talk about a much desired feature of feedback control that may not be immediately apparent: *disturbance rejection*. Consider a modified version of the cruise control problem where the car is on a slope, as illustrated in Fig. 2.12. Newton's law applied to the car produces the differential equation:

$$m \dot{y}(t) + b y(t) = f(t) - mg \sin(\theta(t)),$$

where $y = \dot{x}$ is the velocity of the car, θ is the angle the slope makes with the horizontal, and g is the gravity acceleration⁶. When $\theta = 0$ the car is on the flat, and the model reduces to (2.1). As before, adopting a linear model for the relationship between the force f and the pedal excursion u , $f = pu$ from (2.2), produces the differential equation

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t) - g \sin(\theta(t)).$$

This is a linear differential equation except for the way that $\theta(t)$ enters the equation. In most cases, the signal $\theta(t)$ is not known ahead of time, and can be seen as a *nuisance* or *disturbance*. Instead of working directly with $\theta(y)$ it is conve-

⁶As we insist in using a non-standard unit for measuring velocity (mph), g will have to be expressed in mph/s, or $g = 9.8 \text{ m/s}^2 \approx 9.8 \cdot 3600/1609 \approx 21.9 \text{ mph/s}$.

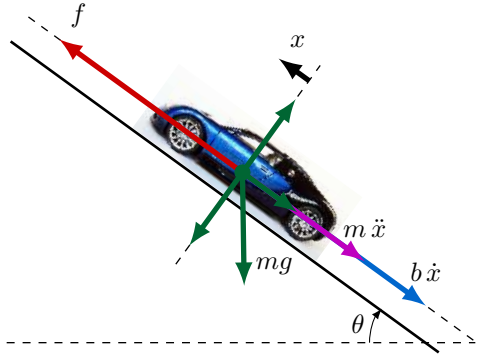


Fig. 2.12: Free-body diagram showing forces acting in a car on a road slope

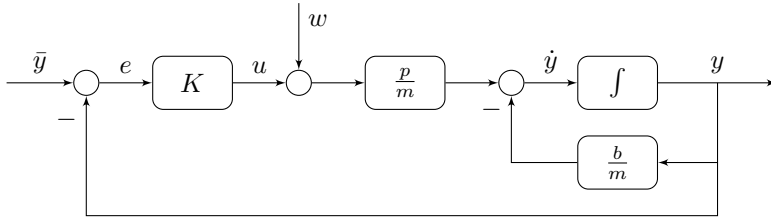


Fig. 2.13: Closed-loop connection of the car showing the slope disturbance $w = -mg/p \sin(\theta)$

nient to view the disturbance signal

$$w(t) = -\frac{mg}{p} \sin(\theta(t)), \quad (2.17)$$

as affecting the linear model

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t) + \frac{p}{m} w(t).$$

This differential equation is completely linear and can be analyzed with simpler tools.

The car model with the input disturbance is represented in closed-loop by the block-diagram in Fig. 2.13, which correspond to equations

$$\dot{y}(t) + \frac{b}{m} y(t) = \frac{p}{m} u(t) + \frac{p}{m} w(t), \quad u(t) = K(\bar{y} - y(t))$$

or, after eliminating u ,

$$\dot{y}(t) + \left(\frac{b}{m} + \frac{p}{m} K \right) y(t) = \frac{p}{m} K \bar{y} + \frac{p}{m} w(t).$$

In order to understand how disturbances affect the closed-loop behavior we analyze the following scenario: suppose that the car is traveling on the flat, $w = \theta = 0$, with the cruise control in closed-loop at the steady-state velocity $y = H(0) \bar{y}$. Under these conditions, the velocity will remain at $y(t) = H(0) \bar{y}$. At time $t = 0$ s, the car hits a 10% grade slope, $\theta \approx 5.7^\circ$, $\bar{w} \approx -0.26$. In order to compute the dynamic response of the car to the slope we set $y(0) = H(0) \bar{y}$, $y(t) = \bar{y}$ and $w(t) = \bar{w}$, $t \geq 0$. It is not hard to see that the dynamic solution $y(t)$ is also given by (2.5) with

$$\lambda = -\frac{b}{m} - \frac{p}{m}K, \quad \tilde{y} = \frac{\frac{p}{m}K\bar{y} + \frac{p}{m}\bar{w}}{\frac{b}{m} + \frac{p}{m}K} = H(0)\bar{y} + D(0)\bar{w},$$

$$D(0) = \frac{p}{b} \frac{1}{1 + \frac{p}{b}K}, \quad (2.18)$$

and

$$\begin{aligned} y(t) &= (H(0)\bar{y} + D(0)\bar{w})(1 - e^{\lambda t}) + H(0)\bar{y}e^{\lambda t}, \\ &= H(0)\bar{y} + (1 - e^{\lambda t})D(0)\bar{w}. \end{aligned}$$

This response is illustrated in Fig. 2.14.

The predicted change in velocity is equal to

$$\Delta y(t) = y(t) - y(0) = y(t) - H(0)\bar{y} = (1 - e^{\lambda t})D(0)\bar{w}.$$

From (2.18), the larger the K , the smaller the change in velocity induced by the disturbance will be. In particular, compare with the change in velocity produced by the open-loop solution which can be shown to be (see also § 1.3.4):

$$\Delta y(t) = (1 - e^{\lambda t})G(0)\bar{w}, \quad G(0) = \frac{p}{b}.$$

Because for any $K > 0$ we have

$$G(0) = \frac{p}{b} > \frac{p}{b} \frac{1}{1 + \frac{p}{b}K} = D(0),$$

we conclude that the feedback solution always provides better *regulation* of the velocity in the presence of a road slope disturbance. Finally, large gains will bring down not only the tracking error but also the *regulation error* in response to a disturbance. Indeed, bigger K 's make both $S(0)$ small and $D(0)$ small.

2.8 Integral Action

We close this chapter analyzing a simple example of a familiar controlled system: a toilet water tank, Fig. 2.15. This system has a property of much interest in control, the so called *integral action*. As seen in previous sections, large

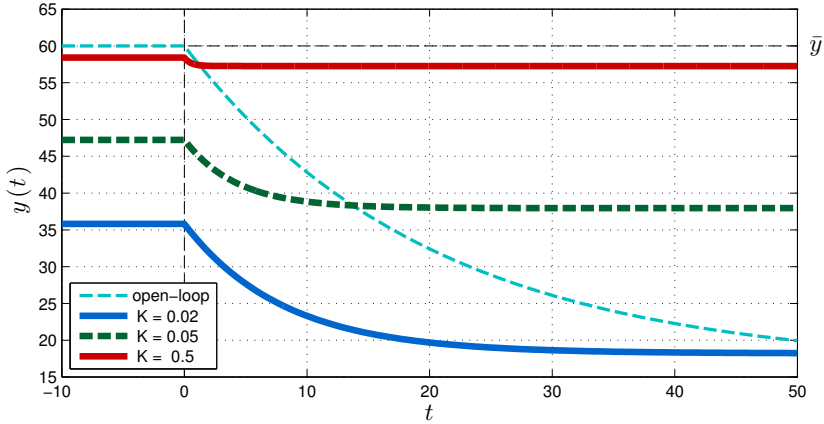


Fig. 2.14: Closed-loop response of the velocity of the car with proportional cruise control (linear model (2.10), $b/m = 0.05$ and $p/b = 73.3$) to a change in road slope at $t = 0$, from flat to 10% grade for various values of control gain

gains will generally lead to small tracking errors but with potential damaging consequences, such as large control inputs that can lead to saturation or other nonlinearities beyond the scope of linear models. In the examples presented so far, only an infinitely large gain could provide *zero* steady-state tracking error in closed-loop. As we will see in this section, integral action will allow closed-loop systems to track constant references with zero steady-state tracking error without resorting to infinite gains.

A schematic of a toilet water tank is shown in Fig. 2.15b. Assuming that the tank has a constant cross-section area, A , the amount of water in the tank, i.e. the volume v , is related to the water level, y , by

$$v = Ay.$$

When the tank is closed, for instance, right after a complete flush, water flows in at a rate $u(t)$, which is controlled by the ballcock valve. Without leaks, the water volume in the tank is preserved, hence

$$\dot{v}(t) = u(t),$$

Combining these two equations in terms of the water level, y , we obtain the differential equation model

$$\dot{y}(t) = \frac{1}{A}u(t) \quad (2.19)$$

which reveals that toilet water tank is essentially a *flow integrator*, as evidenced by its block diagram representation in Fig. 2.16.

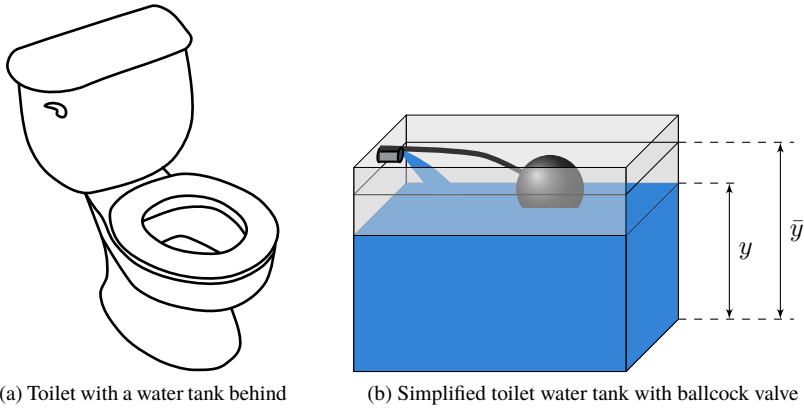


Fig. 2.15: Toilet with water tank and simplified schematic showing the ballcock valve. The tank is in the shape of a rectangular prism with cross section area A . The water level is y and the fill line is $\bar{y}$

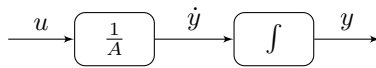


Fig. 2.16: Block-diagram for water tank

A *ballcock valve*, shown in Fig. 2.15b, controls the inflow of water by using a float to measure the water level. When the water level reaches the *fill line*, $\bar{y}$, a lever connected to the float shuts down the valve. When the water level is below the fill line, such as right after a flush, the float descends and actuates the fill valve. This is basically a feedback mechanism. Indeed, we can express the flow valve as a function of the *error* between the fill line, $\bar{y}$, and the current water level, y , through

$$u(t) = K(\bar{y} - y)$$

where the function K behaves as the saturation curves encountered before in Figs. 1.2-1.4. The complete system is represented in the block diagram 2.17, which shows that the valve is indeed a feedback element: the water level, y , tracks the reference level, fill line, $\bar{y}$.

With simplicity in mind, assume that the valve is linear. In this case, the behavior of the tank with the ballcock valve is given by the differential equation:

$$\dot{y}(t) + \frac{K}{A}y(t) = \frac{K}{A}\bar{y}(t).$$

This equation is of the form (2.3) and have a solution (2.5):

$$y(t) = \bar{y}(1 - e^{-(K/A)t}) + y_0 e^{-(K/A)t}, \quad t \geq 0.$$

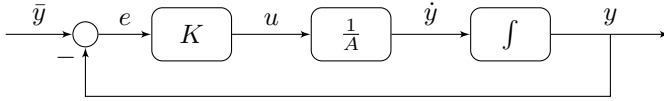


Fig. 2.17: Block-diagram for water tank with ballcock valve

Note the remarkable fact that

$$\lim_{t \rightarrow \infty} y(t) = \bar{y},$$

that is the steady-state solution is always equal to $\bar{y}$, no matter the values of K or A ! The toilet water tank level, y , tracks the fill line level, $\bar{y}$, exactly without a high-gain feedback controller. As it will become clear in the next chapters, the reason for this remarkable property is the presence of a pure integrator in the feedback loop.

Integral action can be understood by looking at the closed-loop diagram in Fig. 2.17. First note that for the output of an integrator to converge to a constant value it is necessary⁷ that its input converge to zero. In Fig. 2.17, it is necessary that $\lim_{t \rightarrow \infty} \dot{y}(t) = 0$. With that in mind, if $y(t)$ converges to anything other than $\bar{y}$, that is $\lim_{t \rightarrow \infty} y(t) = \tilde{y} \neq \bar{y}$ then $\lim_{t \rightarrow \infty} e(t) = \tilde{y} - \bar{y} \neq 0$. But if this is the case, $\lim_{t \rightarrow \infty} (A/K) \dot{y}(t) = \lim_{t \rightarrow \infty} e(t) = \tilde{y} - \bar{y} \neq 0$. Consequently $y(t)$ cannot converge to a constant other than $\bar{y}$. This is true even in the presence of some common nonlinearities in the loop. The ability to track constant references without high gains is the main reason behind the widespread use of integral control. We will analyze integral controllers in more detail later in this book.

We conclude this section by revisiting the car example with linear model (2.3) by noting that when there is no damping, i.e. $b = 0$, then the car becomes a pure integrator. Indeed, in this case

$$\lim_{b \rightarrow 0} H(0) = \lim_{b \rightarrow 0} \frac{\frac{p}{b}K}{1 + \frac{p}{b}K} = 1, \quad \lim_{b \rightarrow 0} S(0) = \lim_{b \rightarrow 0} \frac{1}{1 + \frac{p}{b}K} = 0$$

which implies

$$\lim_{t \rightarrow \infty} y(t) = \tilde{y} = \lim_{b \rightarrow 0} H(0) \bar{y} = \bar{y}$$

independently of the value of K . Large *controller gains* lead not only to small tracking errors but also effective disturbance rejection. The same is true for an integrator *in the controller*, which leads to asymptotic tracking *and* asymptotic disturbance rejection. However, the position of the integrator in the loop matters: an integrator in the system been controlled may lead to zero tracking error

⁷This is a necessary only condition, since $\dot{y}(t) = (1+t)^{-1}$ is such that $\lim_{t \rightarrow \infty} \dot{y}(t) = 0$ but $\lim_{t \rightarrow \infty} \int_0^t \dot{y}(\tau) d\tau = \log(1+t) = \infty$.

but nonzero disturbance rejection error. For instance, in the example of the car we have seen that $b \rightarrow 0$ implies $S(0) \rightarrow 0$ but

$$\lim_{b \rightarrow 0} D(0) = \lim_{b \rightarrow 0} \frac{p}{b} \frac{1}{1 + \frac{p}{b}K} = \frac{1}{K}, \quad \lim_{t \rightarrow \infty} \Delta y(t) = D(0) \bar{w},$$

which gets smaller as K gets large. By contrast, an integrator on the controller will generally lead to both $S(0) = D(0) = 0$ independently of the loop gain K , as we will show later in § 4.4.

Problems

P2.1. Use Newton's Law to show that the first-order ordinary differential equation:

$$m \dot{v} + b v = m g$$

is a simplified description of the motion of an object of mass m dropping vertically under constant gravitational acceleration, g , and linear air resistance, $-b v$.

P2.2. The first-order ordinary differential equation obtained in **P2.1** can be seen as a dynamic system where the output is the vertical velocity, v , and the input is the gravitational force, $m g$. Use § 2.3 to calculate the solution to this equation. Consider $m = 1\text{kg}$ and $b = 10\text{kg s}^{-1}$ and sketch or use MATLAB to plot the response, $v(t)$, when $v(0) = 0$, $v(0) = 1\text{m/s}$, or $v(0) = -1\text{m/s}$.

P2.3. Calculate the position, $x(t)$, corresponding to the velocity, $v(t)$, computed in **P2.2**. Using the same data, sketch or use MATLAB to plot the position $x(t)$, when $x(0) = 0$ and $v(0) = 0$, $v(0) = 1\text{m/s}$, or $v(0) = -1\text{m/s}$.

P2.4. A sky diver weighing 70kg reaches a constant vertical speed of 200km/h during the free-fall phase of the dive and a vertical speed of 20km/h after the parachute is opened. Approximate each phase of the fall by the ordinary differential equation obtained in **P2.1** and estimate the resistance coefficients using the given information. Use $g = 10\text{m/s}^2$. What are the time-constants in each phase? At what time and distance from the ground should the parachute be opened if the landing speed is to be less or equal than 29km/h? If a dive starts at a height of 4km with zero vertical velocity at the moment of the jump and the parachute is opened 60s into the dive, how long is the diver airborne?

P2.5. The first-order nonlinear ordinary differential equation:

$$m \dot{v} + b v^2 = m g, \quad v > 0$$

is a simplified description of the motion of an object of mass m dropping vertically under constant gravitational acceleration, g , and quadratic air resistance, $b v^2$. Verify that:

$$v(t) = \frac{1 + \alpha e^{\lambda t}}{1 - \alpha e^{\lambda t}} \tilde{v}, \quad \tilde{v} = \sqrt{\frac{mg}{b}},$$

$$\alpha = \frac{v_0 - \tilde{v}}{v_0 + \tilde{v}}, \quad \lambda = -\frac{2b\tilde{v}}{m},$$

is a solution to this differential equation.

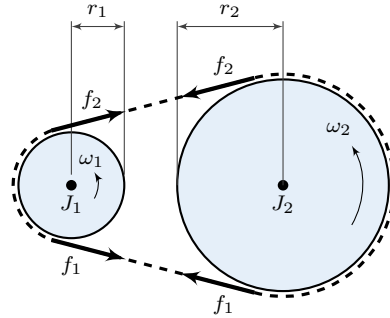


Fig. 2.18: Diagram for **P2.7**

P2.6. Redo problem **P2.4** using the nonlinear model from **P2.5**.

The one-dimensional rotation of a rigid body can be shown to be described by the first-order ordinary differential equation:

$$J \dot{\omega} = \tau$$

where ω is the body's angular speed, J is the body's moment of inertia about its center of mass, and τ is the sum of all torques about the center of mass of the body. In constrained rotational systems, e.g. lever, gears, etc, the center of mass can be replaced by the center of rotation.

P2.7. An (inextensible and massless) belt is used to drive a rotating machine without slip as shown in Fig. 2.18. The simplified motion of the inertia J_1 is described by:

$$J_1 \dot{\omega}_1 = \tau + f_1 r_1 - f_2 r_1$$

where τ is the torque applied by the driving motor and f_1 and f_2 are tensions on the belt. The machine is connected to the inertia J_2 , which is the sum of the inertia of all machine parts. The motion of the inertia J_2 is described by:

$$J_2 \dot{\omega}_2 = f_2 r_2 - f_1 r_2$$

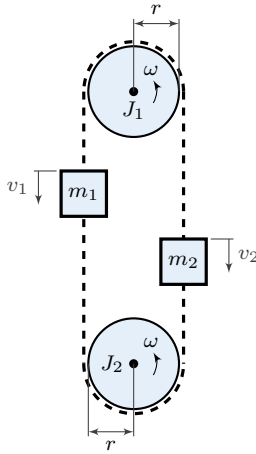
Show that the motion of the entire system can be described by the differential equation:

$$(J_1 r_1^2 + J_2 r_2^2) \dot{\omega}_1 = r_2^2 \tau.$$

and that $\omega_2 = (r_1/r_2) \omega_1$.

P2.8. Why is $f_1 \neq f_2$ in **P2.7**? Under what conditions are f_1 and f_2 equal?

P2.9. Redo problem **P2.7** in the presence of viscous friction torques, $-b_1 \omega_1$ and $-b_2 \omega_2$, on each inertia.

Fig. 2.19: Diagram for **P2.10**

P2.10. A schematic of an elevator is shown in Fig. 2.19. Proceed as in **P2.7** to show that

$$J \dot{\omega} + (b_1 + b_2) \omega = \tau + g r (m_1 - m_2),$$

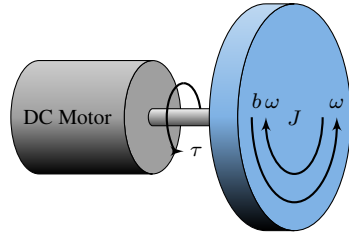
$$J = J_1 + J_2 + r^2 (m_1 + m_2),$$

$v_1 = r \omega$, $v_2 = -r \omega$, is a simplified description of the motion of the entire elevator system where τ is the torque applied by the driving motor on the inertia J_1 , and b_1 and b_2 are viscous friction torque coefficients at the inertias J_1 and J_2 . If m_1 is the load to be lifted and m_2 is a counterweight, explain why it is advantageous to have the counterweight as close as possible to the elevator load.

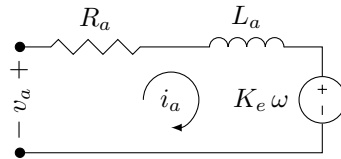
P2.11. Determine a first-order ordinary differential equation based on **P2.10** to describe the elevator as a dynamic system where the output is the vertical velocity of the mass m_1 , v_1 , and the inputs are the motor torque, τ , and the gravitational torque, $g r (m_1 - m_2)$. Use § 2.3 to calculate the solution to this equation. Consider $\tau = 0$, $r = 1\text{m}$, $m_1 = m_2 = 1000\text{kg}$, $b_1 = b_2 = 0$, $J_1 = J_2 = 200\text{kg m}^2$, and $g = 10\text{m/s}^2$. Sketch or use MATLAB to plot the response, $v_1(t)$, when $v_1(0) = 0$, $v_1(0) = 1\text{m/s}$, or $v_1(0) = -1\text{m/s}$.

P2.12. Repeat **P2.11** with $m_2 = 800\text{kg}$.

P2.13. Calculate the (open-loop) motor torque, τ , for the elevator model in **P2.11** so that the vertical velocity of the mass m_1 , v_1 , converges to a desired $\bar{v}_1$ as t gets large.



(a) Mechanical



(b) Electrical

Fig. 2.20: Diagrams for **P2.16**

P2.14. The feedback controller:

$$\tau(t) = K(\bar{v}_1 - v_1(t))$$

can be used to control the ascent and descent speed of the mass m_1 in the elevator discussed in **P2.10** and **P2.11**. As in § 2.5, calculate and solve a differential equation that describes the closed-loop response of the elevator. Using data from **P2.11**, select a controller gain, K , with which the time-constant of the elevator is approximately 1s. Calculate the closed-loop steady-state error between the desired vertical velocity, $\bar{v}_1$, and $v_1(t)$.

P2.15. Repeat problem **P2.14** this time with m_2 as in **P2.10**. As in § 2.7, treat the gravitational torque, $g r (m_1 - m_2)$, as a disturbance. Calculate the closed-loop steady-state error between the desired vertical velocity, $\bar{v}_1$, and $v_1(t)$.

P2.16. The mechanical motion of the rotor of a DC motor shown schematically in Fig. 2.20a can be described by the differential equation:

$$J \dot{\omega} + b \omega = \tau,$$

where ω is the rotor angular speed, J is the rotor moment of inertia, b is the coefficient of viscous damping. The rotor torque, τ , is given by:

$$\tau = K_t i_a$$

where i_a is the armature current and K_t is the motor torque constant. Neglecting the effects of

the armature inductance ($L_a \approx 0$), the current is determined by the circuit in Fig. 2.20b:

$$v_a = R_a i_a + K_e \omega,$$

where v_a is the armature voltage, R_a is the armature resistance, and K_e is the Back-EMF constant. Combine these equations to show that:

$$J \dot{\omega} + \left(b + \frac{K_t K_e}{R_a} \right) \omega = \frac{K_t}{R_a} v_a.$$

This ordinary differential equation can be seen as a dynamic system where the output is the angular velocity, ω , and the input is the armature voltage, v_a . Use § 2.3 to calculate the solution to this equation when v_a is constant. Estimate the parameters of a DC motor that achieves a steady-state angular velocity of 5000RPM when $v_a = 12\text{V}$ and has a time-constant of 0.1s.

P2.17. What other motor parameters could you estimate in **P2.16** if you knew $R_a = 0.2\Omega$? If necessary, propose experiments to determine the remaining parameters.

P2.18. The feedback controller:

$$v_a(t) = K(\bar{\omega} - \omega(t))$$

can be used to regulate the angular speed of the DC motor, $\omega(t)$, for which a model was developed in **P2.16**. As in § 2.5, calculate and solve a differential equation that describes the closed-loop response of the DC motor. Using data from problem **P2.16**, select a controller gain, K , with which the closed-loop steady-state error between the desired angular speed, $\bar{\omega}$, and $\omega(t)$ is less than 10%. Calculate the resulting closed-loop time-constant and the voltage $v_a(0)$ generated in response to a reference $\bar{\omega} = 4000\text{RPM}$ assuming zero initial conditions. What is the maximum value of K for which $v_a(0)$ is smaller than 12V?

The next problems have simple examples of heat and fluid flow using ordinary differential equations. Detailed modeling of such phenomena often requires partial differential equations.

P2.19. A substance's temperature, T in K or in $^{\circ}\text{C}$, flowing in and out of a container kept at the ambient temperature, T_o , with an inflow temperature, T_i , and a heat source, q in W, can be approximated by the differential equation:

$$m c \dot{T} = q + w c (T_i - T) + \frac{1}{R} (T_o - T),$$

where m and c are the substance's mass and specific heat, and R is the overall system's thermal

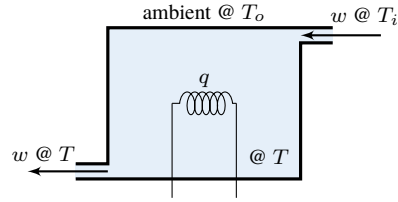
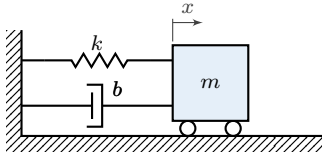
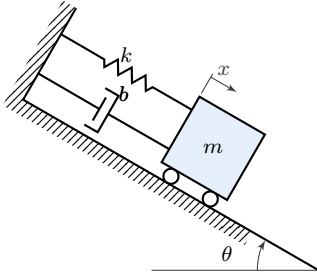


Fig. 2.21: Diagram for **P2.19**

resistance. The input and output flow mass rates are assumed to be equal to w in kg/s. This differential equation model can be seen as a dynamic system where the output is the substance's temperature, T , and the inputs are the heat source, q , the flow rate, w , and the temperatures T_o and T_i . Use § 2.3 to calculate the solution to this equation when q , w , T_o , and T_i are constants.

P2.20. Assume that water's density and specific heat are 997.1kg/m^3 and $c = 4186\text{J/kg K}$. A 50gal ($\approx 0.19\text{m}^3$) water heater is turned off full with water at 140°F ($\approx 60^{\circ}\text{C}$). Use the differential equation in **P2.19** to estimate the heater's thermal resistance, R , knowing that after 7 days left at a constant ambient temperature, 77°F ($\approx 25^{\circ}\text{C}$), without turning it on, $q = 0$, or cycling any water, $w = 0$, the temperature of the water was about 80°F ($\approx 27^{\circ}\text{C}$). In the same conditions, calculate how much time it takes for the water temperature to reach 80°F ($\approx 27^{\circ}\text{C}$) with a constant in/out flow of 2gal/h ($\approx 2.1 \times 10^{-6}\text{m}^3/\text{s}$) at ambient temperature. If the water heater is rated at 40000BTU/h ($\approx 12\text{kW}$) calculate the time it takes to heat up a heater filled up with water from ambient temperature to 140°F ($\approx 60^{\circ}\text{C}$) first without any in/out flow of water, $w = 0$, then with a constant in/out flow of 2gal/h at ambient temperature.

P2.21. Most residential water heaters have a simple on/off type controller: the water heater is turned on at full power when the water temperature, T , falls below a set value, $\underline{T}$, and is turned off when it reaches a second set point, $\bar{T}$. For a heater with the same specifications as in **P2.20** and thermal resistance $R = 0.25\text{m}^2\text{K/W}$, sketch or use MATLAB to plot the temperature of the water during 3 days of a heater filled up with water at ambient temperature, 77°F ($\approx 25^{\circ}\text{C}$), with an on/off controller set with $\underline{T} = 122^{\circ}\text{F}$ ($\approx 50^{\circ}\text{C}$) and $\bar{T} = 140^{\circ}\text{F}$ ($\approx 60^{\circ}\text{C}$), first without any in/out flow of water, $w = 0$, then with a constant in/out flow of 2gal/h at ambient temperature. Compute the average water tem-

Fig. 2.22: Diagram for **P2.23**Fig. 2.23: Diagram for **P2.24**

perature over the entire period and over the last 2 days.

P2.22. Repeat **P2.21** with $\underline{T} = 129.2^\circ\text{F}$ ($\approx 54^\circ\text{C}$) and $\overline{T} = 132.8^\circ\text{F}$ ($\approx 56^\circ\text{C}$). What is the impact of the choice of $\underline{T}$ and $\overline{T}$ on the performance of the controller?

Figs. 2.22 through 2.26 show diagrams of mass-spring-damper systems. Assume that there is no friction between the wheels and the floor and, that all springs and dampers are linear: elongating a linear spring with rest length ℓ_0 by $\Delta\ell$ produces an opposing force $k\Delta\ell$ (Hooke's Law), where $k > 0$ is the spring stiffness; changing the length of a linear damper at a rate $\dot{\ell}$ produces an opposing force $b\dot{\ell}$, where b is the damper's damping coefficient.

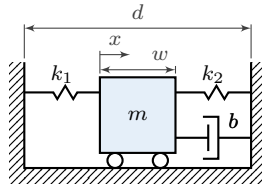
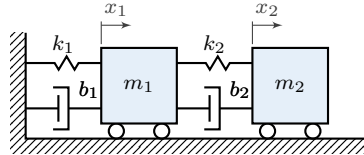
P2.23. Use Newton's Law to show that the second-order ordinary differential equation:

$$m\ddot{x} + b\dot{x} + kx = 0$$

is a simplified description of the motion of the mass-spring-damper system in Fig. 2.22 where the coordinate x is chosen so that $x = 0$ when the length of the spring is equal to its rest length ℓ_0 . Why the equations do not depend on the rest length ℓ_0 ?

P2.24. Use Newton's Law to show that the second-order ordinary differential equation:

$$m\ddot{x} + b\dot{x} + kx = m g \sin \theta$$

Fig. 2.24: Diagram for **P2.26**Fig. 2.25: Diagram for **P2.28**

is a simplified description of the motion of the mass-spring-damper system in Fig. 2.23 where g is the gravitational acceleration and the coordinate x is chosen so that $x = 0$ when the length of the spring is equal to its rest length ℓ_0 .

P2.25. Rewrite the ordinary differential equation obtained in **P2.24** in terms of:

$$y = x - k^{-1} m g \sin \theta$$

Explain the result.

P2.26. Use Newton's Law to show that the second-order ordinary differential equation:

$$m\ddot{x} + b\dot{x} + (k_1 + k_2)x = 0$$

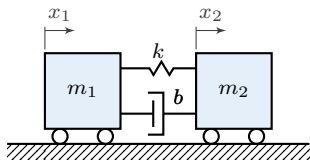
is a simplified description of the motion of the mass-spring-damper system in Fig. 2.24. What is the corresponding choice of x made in this case? Why the equations do not depend on the rest length ℓ_0 nor the dimensions d and w ? What is the difference between the cases $d \geq w + 2\ell_0$ and $d \leq w + 2\ell_0$?

P2.27. Can you replace the two springs in **P2.26** by a single spring and still obtain the same ordinary differential equation?

P2.28. Use Newton's Law to show that the second-order ordinary differential equations:

$$\begin{aligned} m_1 \ddot{x}_1 + b_1 \dot{x}_1 + k_1 x_1 &= 0, \\ m_2 \ddot{x}_2 + b_2 (\dot{x}_2 - \dot{x}_1) + k_2 (x_2 - x_1) &= 0 \end{aligned}$$

constitute a simplified description of the motion of the mass-spring-damper system in Fig. 2.25

Fig. 2.26: Diagram for **P2.29**

where $x_1 = x_2 = 0$ when the length of both springs is their rest length.

P2.29. Use Newton's Law to show that the second-order ordinary differential equations:

$$m_1 \ddot{x}_1 + b(\dot{x}_1 - \dot{x}_2) + k(x_1 - x_2) = 0,$$

$$m_2 \ddot{x}_2 + b(\dot{x}_2 - \dot{x}_1) + k(x_2 - x_1) = 0,$$

constitute a simplified description of the motion of the mass-spring-damper system in Fig. 2.26 where $x_1 = x_2 = 0$ when the length of the spring is its rest length. Show that it is possible to *decouple* the equations if you write them in the coordinates

$$y_1 = x_1 + x_2, \quad y_2 = x_1 - x_2.$$

Explain why using what you know from physics. What happens when $m_1 = m_2$?